

An All-Subgroup Containment-Threshold Atlas for a Frozen Hénon Core

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Abstract

We replace exact full-core repair by a target-containment observable on the twenty actual subgroup rows of $Q = \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2$. For a deletion set D and retained labels $A = L \setminus D$, define $\tau_H(D) = \min\{|R| : R \subseteq D, H \subseteq \Phi(A \cup R)\}$. An exhaustive receipt stores all 20×65536 integer values, their deleted-cardinality tables, and target distributions. The full target row is exactly the C78 repair distance, with distribution 30400, 32704, 2368, 64. An independent target-antichain reconstruction, symbolic marginal checks, clean replay, and thirteen hostile mutations pass. The result is a finite named-support atlas and makes no arithmetic, local, Burnside-ring, table-of-marks, or Hilbert–Polya claim.

1 Targeted repair

Let $L = \{S_1, \dots, S_{16}\}$ be the frozen named presentation. For every subgroup row $H \leq Q$ and deletion set $D \subseteq L$, put

$$\tau_H(D) = \min\{|R| : R \subseteq D, H \subseteq \Phi((L \setminus D) \cup R)\}.$$

This is deliberately a containment threshold. It does not require the generated subgroup to equal H ; in particular, different rows of the same order remain separate named targets.

2 Finite antichain identity

Theorem 1. *For a fixed target H , let $\mathcal{M}_H$ be the inclusion-minimal named supports M satisfying $H \subseteq \Phi(M)$. Then*

$$\tau_H(D) = \min_{M \in \mathcal{M}_H} |M \setminus (L \setminus D)|.$$

Proof sketch. Any feasible restoration contains a target-containing support. Removing labels one at a time while containment persists reaches a member of $\mathcal{M}_H$, and removing labels from the retained part is impossible by the definition of the set difference. Conversely, restoring $M \setminus A$ is feasible for every $M \in \mathcal{M}_H$. $\square$

The C80 checker reconstructs the point-set addition table from C75, derives each $\mathcal{M}_H$, and evaluates this formula for every one of the 65536 retained masks. The twenty target orders in row order are

$$1, 2, 3, 3, 3, 3, 6, 6, 6, 6, 9, 9, 9, 9, 18, 18, 18, 18, 27, 54.$$

3 The exact-core boundary

The last row is $H = Q$. Its threshold values are

$$\#\{D : \tau_Q(D) = 0, 1, 2, 3\} = (30400, 32704, 2368, 64).$$

The source-bound pivot/block decomposition gives, independently,

$$\tau_Q(D) = \mathbf{1}_{S_9 \in D} + \max(0, t(D) - 2),$$

where $t(D)$ counts fully deleted blocks of sizes 1, 1, 2, 5. Thus the C80 row is exactly C78’s ρ row, while the other nineteen rows describe proper-subgroup containment and can have smaller thresholds.

Every deleted-cardinality table has row sum $\binom{16}{k}$, hence each target’s cardinality marginal is $(1+x)^{16}$. If $H_1 \subseteq H_2$, then

$$\tau_{H_1}(D) \leq \tau_{H_2}(D) \quad \text{for every } D,$$

which the receipt checks on all profiles using the actual subgroup bitsets.

4 Certification and scope

The producer binds the C75, C76, and C78 evidence/manifests by raw-byte SHA-256. The canonical C80 evidence hash is `8d27428b14dbd7354e9c8308ad76b1108e3f551702165833301509cd52de7df5`. The independent checker uses target antichains rather than the producer's descending recurrence; a SymPy script verifies all twenty cardinality marginals and the exact C78 polynomial; clean replay preserves the digest; and 13/13 hostile mutations are rejected.

This receipt concerns only the named sixteen-label finite model. It does not claim a full Burnside ring or table of marks, arithmetic/local data, Euler factors, root numbers, automorphy, or a Hilbert–Polya operator.